Progress in Aerospace Sciences 92 (2017) 25–38

Contents lists available at ScienceDirect

Progress in Aerospace Sciences

journal homepage: www.elsevier.com/locate/paerosci

# A review of aircraft turnaround operations and simulations

Michael Schmidt[1]

*Munich Aerospace e.V., Faculty of Aerospace Engineering, Willy-Messerschmitt-Str. 1, Taufkirchen 82024, Germany*

ARTICLE INFO

*Keywords:*
Ground operation
Turnaround
Passenger boarding
Aircraft cabin
Airport

ABSTRACT

The ground operational processes are the connecting element between aircraft en-route operations and airport infrastructure. An efficient aircraft turnaround is an essential component of airline success, especially for regional and short-haul operations. It is imperative that advancements in ground operations, specifically process reliability and passenger comfort, are developed while dealing with increasing passenger traffic in the next years. This paper provides an introduction to aircraft ground operations focusing on the aircraft turnaround and passenger processes. Furthermore, key challenges for current aircraft operators, such as airport capacity constraints, schedule disruptions and the increasing cost pressure, are highlighted. A review of the conducted studies and conceptual work in this field shows pathways for potential process improvements. Promising approaches attempt to reduce apron traffic and parallelize passenger processes and taxiing. The application of boarding strategies and novel cabin layouts focusing on aisle, door and seat, are options to shorten the boarding process inside the cabin. A summary of existing modeling and simulation frameworks give an insight into state-of-the-art assessment capabilities as it concerns advanced concepts. They are the prerequisite to allow a holistic assessment during the early stages of the preliminary aircraft design process and to identify benefits and drawbacks for all involved stakeholders.

## 1. Introduction

The aviation industry will be challenged by an annual 4.6–4.9% growth in passenger traffic in the next 20 years [1,2]. It is imperative that advancements in ground operations, specifically process reliability and passenger comfort, are developed to deal with increasing congestion at major hub airports. Current research mostly focuses on aircraft efficiency in terms of reduced carbon dioxide ($CO_2$), nitrogen oxides ($NO_x$) and noise emissions. This trend is due to ambitious goals promoted by national and international regulators, such as NASA Advanced Air Vehicle Program [3], Air Transport Action Group [4] and the Advisory Council for Aviation Research and Innovation in Europe (ACARE) [5]. However, the implementation of current research in fulfillment of the goals proposed by these regulators would require significant operational efficiency improvements, a topic which is often not addressed. The ACARE work group proposed a reduction of turnaround times by 40% in 2050 using novel handling concepts as well as actual arrival and departure times to be within one minute of scheduled times [6,7]. Especially, regional and short-to-medium haul flights are of concern as they account for the major share of global air traffic.

An efficient aircraft turnaround is an essential component of airline success, especially for regional and short-haul operations. The current ground operational procedures are highly optimized for available infrastructure and aircraft types. Further improvements can be made by refining process execution, providing better predictability for each step improving on-time performance and by reducing additional planned buffer times. In short-haul operations, passenger egress and ingress together with refueling, cleaning, and catering, are on the critical path which determines the total turnaround time. Reducing passenger boarding and disembarking time would simultaneously shorten turnaround time and free up airport capacity. Turnaround time and punctuality are not on the same level of criticality for long-haul operations since delays can be absorbed during the longer flight time and airport curfew hours can determine scheduled arrival and departure times.

The review presented here is based on the research of Schmidt et al. [8–11]. After an introduction of ground operations, focusing on the aircraft turnaround and passenger processes, the key challenges for current aircraft operators are highlighted. These comprise, amongst others, airport capacity constraints, schedule disruptions and the increasing cost pressure for aviation stakeholders. A summary of existing modeling and simulation frameworks gives insight into state-of-the-art assessment capabilities when it concerns advanced concepts. Afterwards, a review of conducted studies and conceptual work focusing on the turnaround shows pathways for potential process

*E-mail address:* mi.schmidt@tum.de.
[1] Scholarship recipient, Munich Aerospace e.V.

http://dx.doi.org/10.1016/j.paerosci.2017.05.002
Received 3 January 2017; Received in revised form 9 May 2017; Accepted 9 May 2017
Available online 23 May 2017
0376-0421/ 

**Nomenclature**

*Acronyms and abbreviations*

| | |
|---|---|
| ABS | Agent-based simulation |
| ACARE | Advisory Council for Aviation Research and Innovation in Europe |
| AHM | Airport Handling Manual |
| APU | Auxiliary power unit |
| CA | Cellular automaton |
| CAST | Comprehensive Airport Simulation Tool |
| $CO_2$ | Carbon dioxide |
| CPM | Critical path method |
| DES | Discrete event simulation |
| CS | Certification Specification |
| DEL | Indira Gandhi International Airport |
| DOC | Direct operating cost |
| EWR | Newark Liberty International Airport |
| FAR | Federal Aviation Regulations |
| FRA | Frankfurt Airport |
| GSE | Ground support equipment |
| HL | Hand luggage |
| IATA | International Air Transport Association |
| IGOM | IATA Ground Operations Manual |
| JFK | John F. Kennedy International Airport |
| LCC | Low cost carrier |
| LHR | London Heathrow Airport |
| LSP | Lifting seat pan |
| MEX | Mexico City International Airport |
| $NO_x$ | Nitrogen oxides |
| OEM | Original equipment manufacturer |
| PCA | Pre-conditioned air |
| PEK | Beijing Capital International Airport |
| SA | Single-aisle |
| SFS | Sideways foldable seat |
| TA | Twin-aisle |
| ULD | Unit load devices |

optimization. Initial results using an agent-based passenger flow simulation are presented for promising aircraft cabin modifications. This paper concludes with a recapitulation of the current research in ground operations and proposes future research goals.

## 2. Overview of current aircraft ground operations

Airports have been repeatedly challenged, in terms of their operations, by the introduction of new aircraft throughout the history of aviation. Some of those aircraft were revolutionary for their time, such as the Boeing 747 (B474), Concorde and recently, the Airbus A380. Before the B747 and Concorde entered service, a thorough understanding of the terminal-related functions, ground handling characteristics and operational economics was necessary for the determination of aircraft servicing so it would not be determined by facility limitations [12]. The B747 revolutionized the cargo handling in terms of speed, through the introduction of further automation with unit load devices (ULD).[2] Prior to the entry into service of the A380, the stakeholders involved analyzed their current airport systems to identify shortcomings and requirements for area improvements [13]. Consequently, airports had to expand their runways, taxiways and gate positions to accommodate this new aircraft type. Applying modified processes in conjunction with adapted ground support equipment (GSE) allowed new aircraft types to be successfully operated at existing airports and partly improve the operation of existing aircraft types. The next generation of aircraft featuring more electric systems, hybrid-electric propulsion or non-drop-in fuels will again challenge the airport operations [14].

This section provides an overview of aircraft ground operations for contemporary single- and twin-aisle single-deck aircraft. The aircraft characteristics related to ground operation are reviewed first, proceeded by the procedures of aircraft turnaround and passenger ingress. Finally, a summary of the regulations and guidelines concerning aircraft design and ground operations currently in place is given.

### 2.1. General overview of aircraft ground operation

Airports are the fundamental core of all commercial passenger flights. They allow aircraft to take-off and land and provide necessary facilities to service the aircraft. Fig. 1 provides an overview of a generic airport. Airport operations are divided into 'landside processes', where passengers arrive, drop off their luggage and go through security, and 'airside processes', where passengers board and disembark planes. Airside processes cover also the take-off and landing of the aircraft, as well as taxiing procedures [15]. The focus here is on the turnaround, the connecting element between the airport and the aircraft. Turnaround can take place directly at the gate or at a remote apron position.

### 2.2. Aircraft characteristics related to ground operation

Aircraft are equipped with multiple interfaces, such as doors and hatches, to exchange goods, passengers or liquids during the ground service. The position of these interfaces varies depending on if the aircraft features wing-mounted or aft-fuselage-mounted engines, or if it is a low-wing or high-wing aircraft. A cluster analysis of regional and single-aisle (SA) aircraft in and out of production [11] revealed a trend shifting towards commonality, such as for the potable and waste water connector at the aft fuselage (see Fig. 2). Also, the electrical power interfaces were in general located in the forward section of the fuselage, while the fuel connector was located on the outer side from the engine. In general, passenger doors were located at each end of the passenger cabin, enabling an uninterrupted passenger cabin where passenger emergency passways do not obstruct the seat rows. This allows for a freely adjustable seating layout in terms of seat pitch[3] and class configuration. Galleys and lavatories were situated in the entrance area, enabling an easy trolley exchange through the opposite service door during the aircraft turnaround. Larger SA aircraft featured additional quarter doors in front of the wing due to their higher passenger capacity which allow for passenger egress and ingress. Due to the spatial separation of passenger boarding and cargo operations, cargo doors were, in general, located on the right side. High-wing configurations with turboprop engines allow boarding and disembarking as well as loading and unloading operations to be easily performed without the requirement of external airstairs due to the lower sill height [11].

The seating layout inside the fuselage is driven by the number of passengers transported, door positions and the operator business model. The latter determines the number of cabin classes, the targeted aircraft operation in terms of average flight distance and the seat pitch. The existence of premium cabin classes requires a higher ratio of lavatories and galleys per passenger compared to SA layouts with only

[2] A unit load device (ULD) is a standardized pallet or container used to load luggage or freight on to an aircraft.

[3] The seat pitch is defined as the distance from any point on one seat to the same point on the following seat.

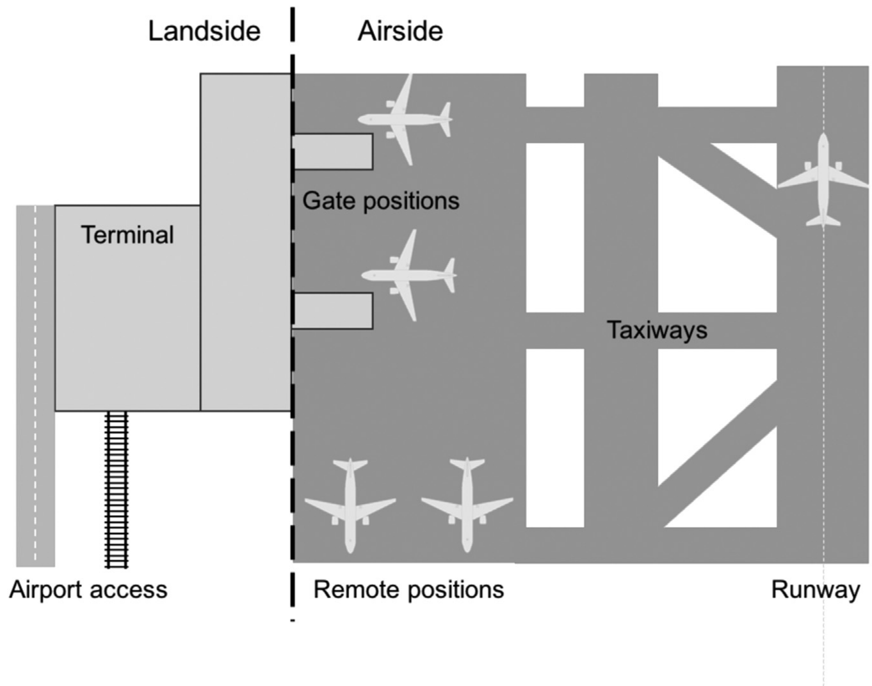


**Fig. 1.** Generic airport with landside and airside elements (based on [15]).

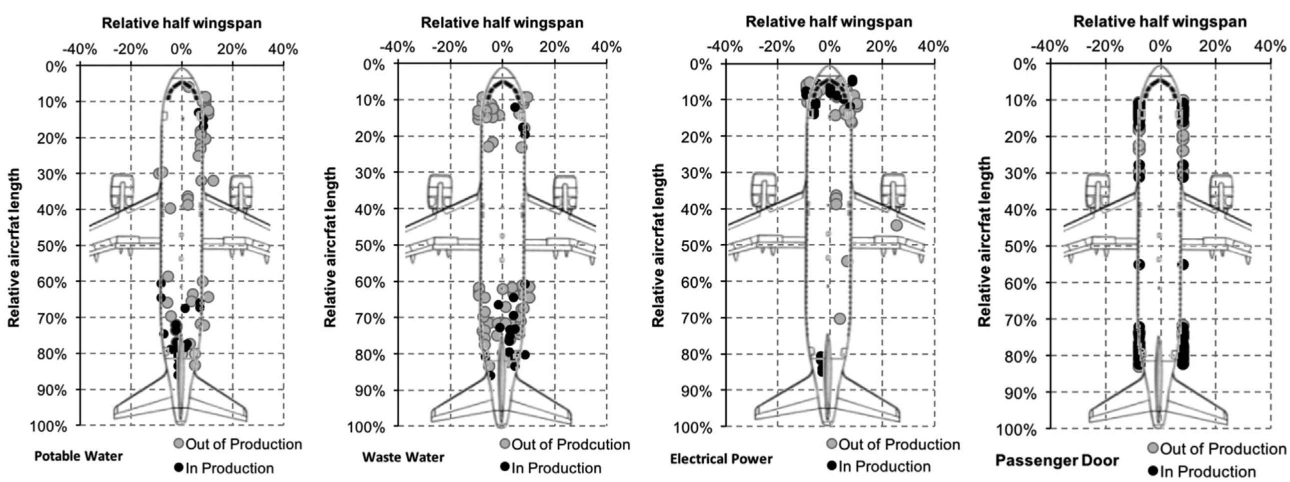


**Fig. 2.** Aircraft interface locations for short-to-medium haul aircraft (adapted from [11]).

economy seats. This applies also for long-haul operations. Overhead bins for luggage storage are mounted on the ceiling along the aisles. For SA aircraft above 50 passengers, a common seat abreast varies between 4–6 and for twin-aisle (TA) aircraft it ranges from 7 to 10. The resulting fuselage width should be compatible to fit current ULD types into the cargo compartment.

### 2.3. *Aircraft turnaround process*

Aircraft ground handling, or also referred to as turnaround, is a fundamental part of commercial aircraft operations and describes all operations for preparing an aircraft for flight. The turnaround process starts when the aircraft reaches the parking position after landing and the chocks are set ('on-block time'). The parking position can be either located at the terminal, referred to as gate position, or on the apron, known as remote position (see Fig. 1). The process ends when the aircraft is ready to leave and the chocks are removed ('off-block time'). Commercial aircraft depend on vehicles and systems known as GSE to perform the required processes. In general, the turnaround time depends on the aircraft type, the number of passengers, amount of loaded and unloaded cargo, as well as, the business model of the aircraft operator. The aircraft turnaround consists of various processes which are, in general, provided by more than one service provider. The course of activities follows a strict chronological order, however some processes can be executed concurrently, while others only sequentially.

The ground power supply is connected after the chocks are placed before the wheels. This allows the engines and auxiliary power unit (APU) to be turned off. If the climate conditions require, the pre-conditioned air (PCA) unit is connected. A passenger boarding bridge typically docks at the front left side door at the terminal parking positions. On remote apron positions, passenger stairs or aircraft-integrated stairs are used on the forward and aft left side door. As soon as the doors are opened, passenger disembarking begins and simultaneously cargo and baggage are unloaded. Also at this time, the potable water is replenished. Hygienic standards prescribe this service to be completed before waste water servicing may commence (see Airport Handling Manual (AHM) 440 [16]). The equipment may be repositioned, since certain aircraft feature more than one waste tank. An aircraft is refueled after the last passenger has left the aircraft, according to requirements stated in EU-OPS 1.305 (FAR 121.570) [17,18]. In the meantime, the flight crew can begin preparation for the next flight, check the airworthiness of the aircraft with a walk around, setup the flight computers, and execute system checks. The cabin crew examines the general cabin condition and cabin emergency equipment [19]. Inside the aircraft cabin, the catering provider exchanges the trolleys and the cabin interior is cleaned and prepared for the next flight. These operations are usually performed in the absence of passengers because of noise and comfort issues. Once the cargo and baggage unloading is complete, the loading for the next flight can commence. After the fuel has been completely replenished, the passenger ingress is initiated and a final head count is performed before leaving the parking position. Electrical power switches from the ground power supply to the APU. The chocks are removed and, if the aircraft is parked at nose-in position, a pushback is required [20].

Fig. 3 depicts a typical top view of the ramp layout at the gate position for a short-to-medium haul aircraft. The left side doors are used for the passenger egress and ingress and the right doors are utilized for catering and cargo handling. Commonly, the position of the service vehicles is predefined due to the interface locations of the aircraft. The corresponding Gantt-chart (see Fig. 4) presents the duration of individual handling operations in a sequence. The logical chain, regulations and restrictions due to limited space around the aircraft leads to a strict chronological order for certain handling operations. These operations mark the critical path of the turnaround process, since the minimum necessary turnaround time depends on these. In most instances, the critical path consists of the passenger and aircraft cabin activities, however in several circumstances, the fueling operation may become the critical path. Other activities, such as unloading, loading and aircraft servicing, can normally be performed without impact on or from the critical path activities (AHM 021) [16]. Analysis for handling processes of short-haul flights by Fricke and Schultz [21] underline that besides passenger processes, catering and refueling also have a major impact on the overall turnaround time. Thus, reducing time devoted to these processes would shorten turnaround time, which in turn, directly influences the gate utilization and the number of flights that an aircraft can perform by day.

In the SA aircraft segment between 100 and 200 passengers, the turnaround time averages 35 min (with a maximum of 51 min), as illustrated in Fig. 5. The required time averages around 17 min for regional aircraft and 61 min for TA aircraft. However, the actual turnaround time of an aircraft is of a stochastic nature [22] since passenger numbers, replenished fuel, and cargo loads, vary from flight to flight. Airlines try to handle this variance through the incorporation of buffer times which leads to a large variation of scheduled on-block times compared to the original equipment manufacturer (OEM) guidelines.

### 2.4. Passenger egress and ingress

As stated by Marelli et al. [23], the passenger aircraft boarding process has been an issue since the late 1970's. Over the past decades, the average boarding velocity dropped from around 20 passengers per minute to nearly nine passengers per minute. This decline is a result from increased hand luggage (HL), diverse airline service strategies based on passenger boarding schemes, and changing passenger demographics in terms of their composition and behavior. Other factors which influence the process include aircraft configuration, cabin layout, boarding schemes, passenger anthropometrics and airport environment. One contributing aspect that cannot be influenced by the aviation stakeholders are the passengers themselves. The passenger ratio of male and female air travelers changes, as well as their anthropometrics in terms of average weight and body size [24–27].

Common pre-boarding practices sequence families with small children and passengers with restricted mobility to board the aircraft first. Second, passengers booked in premium travel classes or with frequent flyer status are allowed to board the aircraft and finally all other remaining passengers. After passengers have entered the aircraft, their initial goal is to search for their assigned seat. They are limited in their movements due to the constricted aisle layout. Prior to taking a seat, the outerwear is taken off and HL is stowed under the seat or in the overhead bins. If a passenger blocks the aisle during this time, the ‘stowing task’, it is referred to as ‘aisle interference’, as illustrated in

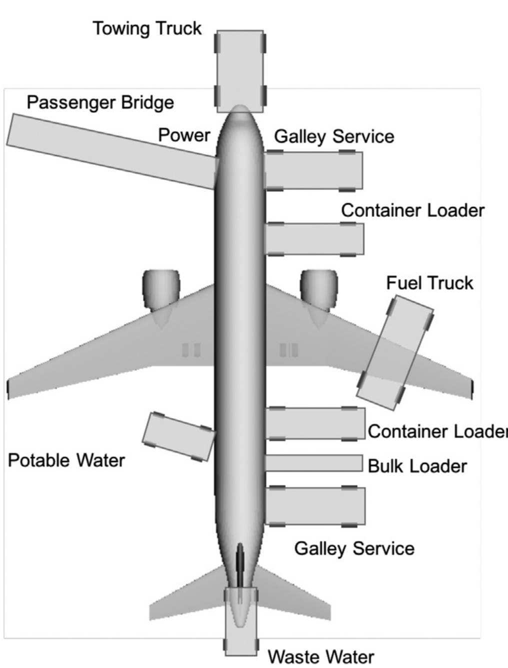


**Fig. 3.** Typical ramp layout at a gate position for a single-aisle aircraft (based on [20]).

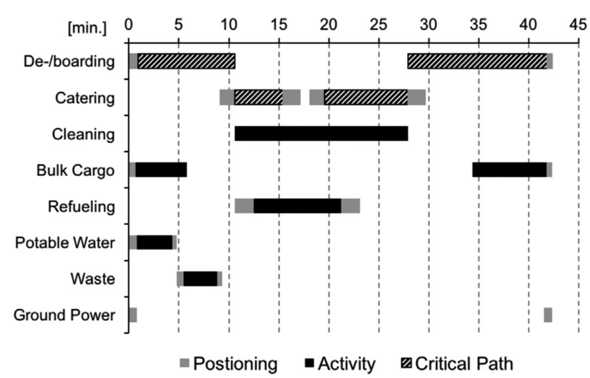


**Fig. 4.** Turnaround Gantt-chart for a typical single-aisle aircraft (based on [20]).

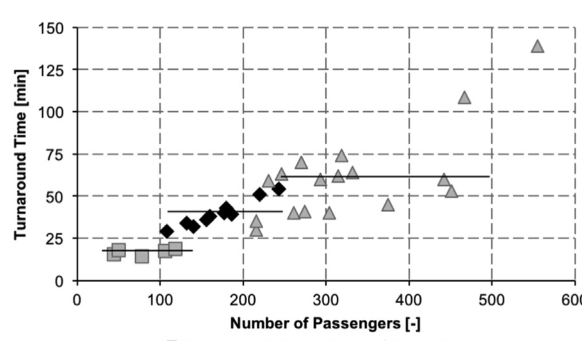


**Fig. 5.** Turnaround time correlation with the number of passengers for regional, single-aisle and twin-aisle aircraft (based on manufacturer data).

Fig. 6. A 'row interference' occurs when a passenger must wait for another passenger in their row to sit down before they can enter the row, this often also results in an aisle interference.

Passengers disembarking behavior, although not organized or planned, often follows a distinct pattern; passengers will wait for other passengers in the rows before them to leave before leaving themselves. This causes a queuing of passengers in the back rows. In most cases, passengers in the same row will leave from aisle seat to window seat eliminating the row interferences. The passenger's behavior in an aircraft is comparable to a crowd behavior where everyone tries to get to the same destination first [28].

In general, methods for a shorter overall boarding times offer a reduced ingress time for each individual passenger [29]. Hence, a smooth boarding procedure would benefit passengers and airlines in the same way, since the boarding procedure is the last process to complete the turnaround [30].

### 2.5. Regulations and guidelines in place

The regulatory framework for turnaround operations examines the procedures from an aircraft design point of view in the Certification Specification (CS) for large airplanes CS-25 [17] and Federal Aviation Regulations (FAR) part 25 - airworthiness standards: transport category airplanes [18]. From an operational perspective, guidelines can be found in the International Air Transport Association (IATA) AHM, IATA Ground Operations Manual (IGOM) [16], as well as the European Commission Regulations [31].

An aircraft should be staffed with one cabin crew member for every 50 passenger seats installed (EU-OPS 1.990) to ensure safe operation. The crew must be on board if passengers are in the cabin during ground operations (EU-OPS 1.311). While passengers board or disembark, the aircraft should not be refueled. Otherwise necessary precautions must be taken to initiate a required evacuation of the aircraft (EU-OPS 1.305, AHM 175, AHM 630). Before the departure, all hand baggage which is taken into the passenger cabin should be adequately and securely stowed (EU-OPS 1.270).

The IATA AHM [16] covers process specifications for passenger, baggage and cargo handling, and moreover serves as a guideline for aircraft loading and airside safety management. All aircraft operators should determine a ground time which is designed to meet operational requirements, but should not compromise safety (AHM 021). In general, the potable water servicing should comply with the World Health Organization standards and the connectors shall be kept a certain distance away from the waste storage or treatment and toilet servicing equipment (AHM 440, AHM 441). The IGOM delivers step-by-step procedures for each necessary handling task. Cabin baggage cannot be accepted if it is unsuitable due to its weight, nature, or does not fit under the seat or in the overhead compartments. Baggage is characterized as bulky or oversized if its weight exceeds 32 kg (70 lb). Also, the incident reporting duty is highlighted, since aircraft damage can endanger passengers and employees, and the resulting disruptions could negatively impact safe airline operations [16].

## 3. Current ground-operational challenges for aircraft operators

Ground operational processes are the connecting element between aircraft en-route operations and airport infrastructure, resulting in various dependencies like airport capacity constraints, aircraft type diversity and schedule disruptions. In the following, key challenges for current aircraft operators are highlighted including prolonged passenger process times, airport capacity constraints, schedule disruptions, reduced aircraft utilization and the increasing cost pressure for aviation stakeholders.

### 3.1. Prolonged passenger egress and ingress processes

The use of larger aircraft is one solution to cope with an increase in air traffic, other than increasing the flight frequency. Since the 1960's, the average number of installed seats per aircraft on short-haul segments has increased from 110 to 160 [32]. This is caused by the trend towards denser aircraft cabins, which was driven by low cost carriers (LCC) and by the choice of larger SA aircraft types, such as the A321 or B737-900. Recent aircraft type upgrades for the A320 and B737 family manage to accommodate an increased passenger number while keeping the overall dimensions constant [33,34]. Denser aircraft cabins are an effective development direction, since 10% more passengers in current short-to-medium haul aircraft reduce the fuel burn per passenger mile by 6% [35]. Unfortunately, passengers have to face a reduced level of comfort due to a shrinking seat pitch, reduced catering and a reduced number of lavatories.

Further contributing factors to prolong passenger egress and ingress times are continuously increasing aircraft load factors from 67% in 1995 up to 84% in 2015 [36] together with a trend towards an excessive amount of cabin luggage. This results in longer gate occupation times and could lead to schedule disruptions due to the unpre-

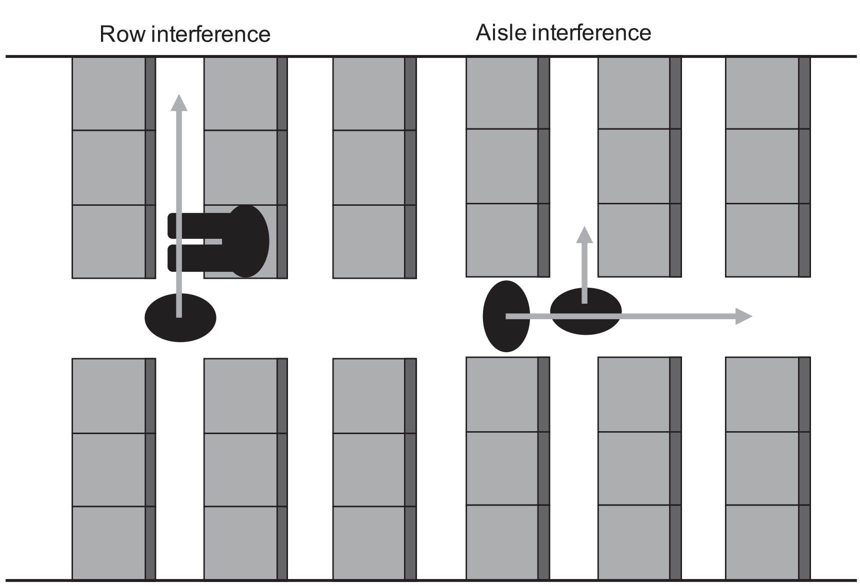


**Fig. 6.** Illustration of row and aisle interferences in the cabin.

dictable passenger behavior.

### 3.2. Capacity constraints at major hub airports

A recent study for European airports [37] revealed that, especially large hub airports handling 15% of the total passengers, are congested and already operate at their capacity limit. These airports, such as London Heathrow (LHR) and Frankfurt (FRA) airport, are constrained by existing infrastructure and are unable to expand due to local and political conditions, lack of available areas and high investments. The remaining 90% of the airports in this study are mainly characterized by moderate to low capacity utilization. The same trend is recognizable around the globe where a multiplicity of hub airports operate close to their maximum capacity and exhibit increasing average delays. It is expected that the challenges for capacity increase of the airport system in certain regions of the world will not change substantially in the near future [37].

### 3.3. Schedule disruptions

During peak hours, congested hub airports often operate at their maximum runway capacity, taxiway and gate utilization. Each schedule disruption due to late arrival of aircraft or unreliable and inefficient ground operation processes could result in significant delays, opening the potential for significant airport operation impairment. The average taxi-out times at airports is an indicator of the congestion level in terms of queue lengths. It exceeds the unimpeded taxi times at large international airports [38]. The taxi-out time averaged between 10 and 20 min in European airports in 2013, with most large hub airports ranging closer to the upper end of this average. For some of the busiest international airports, such as Beijing (PEK), Delhi (DEL) and Mexico City (MEX), taxi-out times average more than 20 min, and in the case of New York (John F. Kennedy International Airport (JFK), Newark Liberty International Airport (EWR)), up to an average of 40 min [39].

High ground congestion levels can lead to schedule disruptions and network delays. Around 45% of the European delays in 2013 were reactionary delays caused by late arrival of aircraft or crew from another flight. A considerable amount of 17% was caused by ground operation disruptions including aircraft and ramp handling, passenger and baggage handling, aircraft damage, and flight operation and crewing, as illustrated in Fig. 7. Technical aircraft failure, weather, authorities and air traffic flow cause the remaining 38% [39]. In general, delays can be partly compensated by means of improved ground process efficiency, scheduled buffer time between two flights or they can be completely absorbed when the aircraft goes off rotation. Furthermore, they can be absorbed during flight on long-haul trips [40].

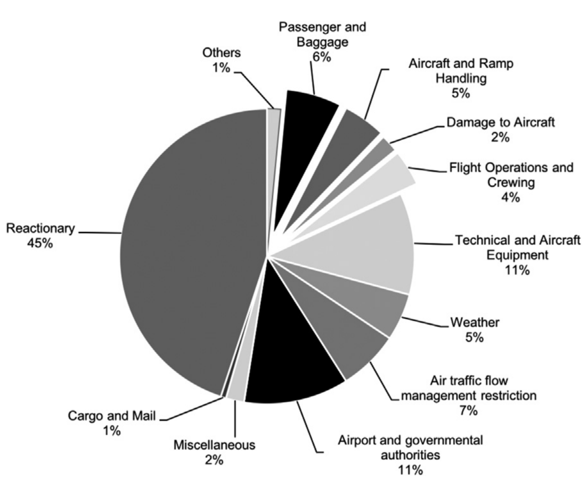


**Fig. 7.** Delay cause in 2013 at European airports [39].

### 3.4. Reduced aircraft utilization

Aircraft utilization measures operational efficiencies as a function of aircraft characteristics, aircraft availability, trip distances and ground time. Improved aircraft utilization allows for the spread of fixed cost of ownership, which accounts for 20% of the direct operating costs (DOC) for current short-to-medium haul aircraft [41], across more trips and passengers resulting in lower cost per available seat mile. For regional and short-haul operations with shorter trips, this increases the number of annual trips, as well as, executed aircraft turnaround events. A reduction from 40 to 30 min turnaround times could increase the number of annual trips by 8% on a 500 nm (926 km) trip [42]. Current LCC try to maximize the airtime of their aircraft through faster ground processes, which notably worked for the airline Southwest, who achieved a low of a 15 min turnaround time. A faster turnaround improves not only the aircraft utilization but also all factors of production such as gates, ground equipment and labor. These essential operational efficiencies result in lower cost which stem from reduced process complexity [43].

### 3.5. Cost pressure for aviation stakeholders

Even if the global airline industry continues to grow rapidly, the ability to deliver cost efficiencies and productivity improvements remain crucial for aviation stakeholders and airlines. Most of the players in the value chain turn a respectable profit, such as providers of global distribution systems, aircraft manufacturer or leasing companies, whereas airlines struggle to break even [44]. This is driven by a significant degree of regulation and the vulnerability to exogenous events, such as security concerns, volcanic eruptions and infectious diseases, and underlines the idea of a worldwide aviation cycle. The resulting price pressure has led to a consistent price fall in airline yields since the 1950s [45]. In this time, the inflation-adjusted yields have also fallen at an annual rate of 3%. Based on productivity improvements in aircraft equipment and operation, airlines were able to lower fares in an effort to remain competitive. The industry is highly cyclical but even in good times does not generate exceptional returns [44].

The emergence and rapid growth of LCC since the 1960's has increased competition within the airline industry and poses significant competitive challenges to network airlines. A sharp rise in jet fuel prices between 2003 and 2008 increased the urgency for overall cost reductions. In 2004, a 36% cost gap in terms of operating costs per available seat kilometer existed for the three largest US network airlines versus Southwest. In Europe, the gap between network airlines and LCCs was even higher with 40–60% and in Asia and Latin America, up to 60–70%. Part of the cost gap reflects the premium service offered by network airlines and their network structure. Other main cost efficiency contributors are product, distribution and overhead, increased seat density, used infrastructure, aircraft, fuel and labor. In matters of ground operations, efficiency gains are envisaged through a standardization of service level agreements and increased crew involvement for cleaning and loading procedures. These gains are available to network airlines and in many cases are already implemented by the LCC [46].

## 4. Review of existing modeling and simulation frameworks for aircraft ground operation

A holistic performance assessment is aspired to during the early stages of conceptual aircraft design projects to claim the required resources for further development in terms of their technology readiness level. This section provides an overview of existing aircraft turnaround modeling approaches, as well as, simulations for passenger egress and ingress.

**Table 1**
Overview of existing turnaround modeling approaches (n.a. = not accessible or currently not available for purchase).

| Name | Year | Methodology | Purpose | Availability | Ref. |
|---|---|---|---|---|---|
| Voulgarellis | 2005 | Matlab | Flight plan | n.a. | [53] |
| Ioannidis | 2005 | Simulink | Flight plan | n.a. | [54] |
| Ip | 2010 | ABS | GSE allocation | n.a. | [51] |
| Norin (Arena) | 2012 | DES | Turnaround | n.a./commercial | [60,61] |
| Vidosavljevic and Tosic | 2016 | Petri nets | GSE allocation | n.a. | [52] |
| Andersson | 2000 | integer programming model | Turnaround | n.a. | [55,56] |
| Tian | 2013 | virtual simulation | Turnaround | n.a. | [64] |
| Mota | 2015 | DES | Turnaround | n.a. | [61] |
| Wu and Caves | 2004–08 | Markovian | TA uncertainties | n.a. | [24,57,58] |
| Fricke et al. | 2008–10 | statistical density functions | TA uncertainties | n.a. | [67,68] |
| Schlegel | 2010 | stochastic functions | TA uncertainties | n.a. | [69] |
| Bevilacqua et al. (ProSim, WITNESS) | 2015 | stochastic functions | Turnaround | n.a./commercial | [64–66] |
| CAST GroundHandling | 2009 | ABS | Turnaround | n.a./commercial | [63] |
| Sanchez | 2009 | stochastic functions | Turnaround | n.a. | [70] |
| Schmidt et al. | 2016 | stochastic functions | Turnaround | n.a. | [14] |

### 4.1. Aircraft turnaround modeling

Various approaches exist to model airport ground operations in general. Most existing frameworks follow the critical path method (CPM) which identifies time constraint aircraft turnaround activities. These models usually do not capture the uncertainties of schedule punctuality or the operational uncertainties of aircraft turnaround operations [47]. The applied modeling techniques cover stochastic probability functions based on historic data, discrete event simulation (DES) and agent-based simulation (ABS) approaches. A summary of the reviewed frameworks is compiled in Table 1.

Focusing on a more detailed perspective, microscopic models are more flexible and represent the entities found in the real system with their individual attributes and behavior. One approach applied to microscopic system modeling are ABS models. These are characterized by common actions of autonomously deciding agents [48]. The system behavior in ABS is modeled as a collection of autonomous decision-making agents, although each agent individually assesses its situation and makes decisions based on a defined set of rules. The modeling from the involved agents' point of view is referred to as individual perspective. The resulting interactions are heterogeneous and can generate network effects [49]. The behavior of the overall system is based on the interactions of the entities as macro phenomenon. This view also characterizes other modeling styles such as individual-based, object-oriented or DES [48,50].

Ip et al. [51] proposed an agent-based model to support the decision making in GSE allocation in an effort to minimize the chance of delays and operating cost. The considered GSE and aircraft were, performance-wise, identical and all process times were assumed to be determinable and known in advance. An approach using Petri nets[4] was pursued by Vidosavljević and Tosić [52] encompassing taxi-in, turnaround and taxi-out procedures. Based on traffic data from Belgrade airport, the number of required GSE could be determined. Voulgarellis et al. [53] proposed an airport ground handling simulation using Matlab that aimed to estimate the number of required GSE to serve a specific flight plan. Using model blocks for each operation, the airport operation dynamics could be simulated and time delays could be predicted. The turnaround itself was not modeled in detailed and was assumed to last for a fixed time period. Following a similar approach, Ioannidis et al. [54] built a model using Simulink. Andersson et al. [55,56] proposed an integer programming model to capture the dynamics of the aircraft turnaround based on available ground operational data. Their goal was to simulate airline operational decisions about pushback times under resource constraints. Wu and Caves [22,57,58] published various studies on the modeling and simulation of aircraft turnaround operations focusing on operational uncertainties. They used a Markovian[5] type simulation model [57] combined with Monte Carlo techniques in order to capture the stochastic effects of flight punctuality and operational uncertainties. With this method, the stochastic transition behavior between major aircraft turnaround activities and potential disruption activities from passengers and aircraft ground services can be modeled. A second approach [58] applied an analytical model to simulate the efficiency of aircraft turnaround operations at airports targeted at minimizing the aviation system costs by balancing trade-offs between schedule punctuality and aircraft utilization.

Focusing more on the individual ground handling processes, a detailed DES model of the turnaround operations at Lelystad airport was developed by Mota et al. [59]. The modular model covered a detailed airport layout representation and performed each ground handling task using fixed process time values. The focus of the work from Norin et al. [60] was on the de-icing services at Stockholm Arlanda airport. The compiled turnaround model covered all procedures from the touch down and taxiing into the stand, the turnaround process itself and ends with taxiing out to the runway and taking off. To model each process, the commercial DES package ARENA [61] was used. The turnaround model included some simplifications in terms of process dependencies. A sophisticated virtual simulation-based approach was presented by Tian et al. [62] to evaluate the handling of advanced aircraft concepts with unconventional configurations, such as blended wing body and joined wing. This framework requires a three-dimensional aircraft model as input. During the simulation, vehicle movement paths are calculated and the user can optimize the proposed arrangement. The commercial software Comprehensive Airport Simulation Tool (CAST) Ground Handling [63] developed by the Airport Research Center comprises ABS ground handling simulation, three-dimensional visualization and collision detection. It provides a chronological ground handling simulation considering rules and delays, compatibility analysis and detailed movement animation. Bevilacqua et al. [64] applied the commercial simulation software ProSim [65] and WITNESS [66] to model ground handling operations. The time for the sub-processes was determined with stochastic functions based on triangular and lognormal distributions using historical data.

Based on an extensive dataset of recorded real turnaround events, Fricke et al. [21,67,68] produced statistical density functions for each sub-process of a turnaround event. However, no distinction of the

[4] Petri nets are a graphical and mathematical modeling technique which consists of places, transitions and arcs.

[5] A Markov model is a part of probability theory used in stochastic modeling to model randomly changing systems where it is assumed that future states depend only on the current state not on the events that occurred before it.

**Table 2**
Overview of existing passenger flow simulation frameworks (n.a. = not accessible or currently not available for purchase).

| Name | Year | Method | Purpose | Availability | Reference |
|---|---|---|---|---|---|
| Landeghem and Beuselinck (Arena) | 2002 | DES | Boarding strategies | commercial | [61,77] |
| Ferrari and Nagel | 2005 | CA | Boarding strategies | n.a. | [78] |
| Livermore et al. | 2008 | ABS | Boarding strategies | open-source | [79] |
| Audenaert et al. | 2009 | ABS | Boarding strategies | n.a. | [80] |
| Steiner and Philipp | 2009 | DES | Boarding strategies | n.a. | [85] |
| Cimler et al. (NetLogo) | 2013 | DES | Boarding strategies | Based on open-source framework | [81,82] |
| Qiang et al. | 2014 | CA | Boarding strategies | n.a. | [83] |
| Schultz | 2010 | CA | Airport terminal | n.a. | [84] |
| TOMICS - DLR | 2011 | DES | Airport terminal, aircraft de-/boarding | n.a. | [87] |
| PEDS - Boeing | 1998 | DES | Aircraft de-/boarding | n.a. | [23] |
| MASim - Richter | 2007 | ABS | Aircraft de-/boarding | n.a. | [88] |
| CAST Cabin - Airport Research Center | – | ABS | Aircraft de-/boarding | commercial | [63] |
| Boeing | 2013 | CA | Aircraft de-/boarding | n.a. | [89] |
| Schultz | 2013 | ABS | Aircraft de-/boarding | n.a. | [30] |
| Fuchte | 2014 | ABS | Aircraft de-/boarding | n.a. | [32] |
| airExodus - Fire SafetyEngineering Group | 2015 | ABS | Aircraft evacuation | commercial | [96] |
| PAXelerate – Schmidt et al. | 2016 | ABS | Aircraft de-/boarding | open-source | [8–10,90] |

aircraft type or operation was conducted. Schlegel [69] pursued a similar approach modeling stochastic functions for the turnaround processes based on recorded flight data. He combined the determined functions to a simulation model analyzing impacts of operational impairments. Sanchez [70] created statistical equations based on real turnaround observations for each individual process of a turnaround. The process times are calculated using a specific flow rate per minute for the exchanged goods, such as containers or passengers. The approach was later picked up by Schmidt et al. [14] and amended with a Monte Carlo approach for determining positioning and removal times.

### 4.2. Passenger egress and ingress simulation

Different approaches to model passenger ingress and egress processes have been proposed in literature, but distinguishing between these concepts is not straightforward. Macroscopic models use a system of differential equations describing associations between variables at the overall system level. A survey showed diverse macroscopic approaches ranging from a physical-mathematical point of view using Lorentzian space-time geometry and a CPM [71,72], a non-linear assignment model with quadratic and cubic terms [73,74] to mixed integer linear programs [75] in combination with genetic algorithms [76].

Table 2 provides an overview of existing microscopic passenger flow model approaches focusing on aircraft and terminal applications. Microscopic models focus on a more detailed modeling perspective, as introduced in Section 4.1.

Dealing primarily with boarding strategies, Van Landeghem and Beuselinck [77] applied the commercial software Arena [62] to investigate boarding strategies. Ferrari and Nagel [78] conducted a computer simulation similar to the latter using a simplified cell-based simulation architecture. An ABS approach was pursued by Livermore [79] and Audenaert et al. [80] to investigate the applicability to assess boarding strategies. Cimler et al. [81,105] used an ABS model based on NetLogo [82] to compare aircraft boarding and disembarking methods. The model enabled the experimentation of various scenarios and parameters settings, such as the number of passengers, the ratio of passengers carrying luggage and the size of the plane. NetLogo[6] is an open-source programmable modeling environment, well suited for modeling complex systems developing over time. Also focusing on boarding strategies, Qiang et al. [83] used a cellular automaton (CA) model based on the Nagel-Schreckenberg model, which is a theoretical model for the simulation of freeway traffic, to test the behavior of boarding strategies under different conditions considering seat and aisle interferences.

A wider range of applications was targeted by the approach of Schultz [84] who developed a stochastic model for passenger movement behavior in airport terminal facilities representing the extension of a spatially discrete microscopic model based on CA. The Airplane Boarding Simulator by Steiner and Philipp [85] used a DES to study the influence of differencing factors on the boarding time. AirExodus [86] is a comprehensive ABS framework especially designed for aircraft evacuation certification which is commercially available. It has the ability to represent agent interaction with signage and smoke, and allows the simulation of thousands of agents without performance penalties.

Focusing on the application to aircraft cabins, Boeing's [23] approach used DES to combine the effects of mathematical queuing theory with an analysis of random behavior. This simulation featured various interior configurations, passenger mixes, and boarding scenarios. Passengers can be assigned certain attributes, such as walking speed, type of carry-on luggage, luggage put-away time, and relationship with other passengers. Schultz et al. [30] proposed a microscopic approach of modeling the passenger flow using an asymmetric simple exclusion process where the passenger was defined as a one dimensional, stochastic time and space discrete transition process. Based on a grid with a rather limited resolution, luggage stowing and seat interferences can be simulated. Other potential disturbances, such as congested overhead bins, missed rows or overtaking passengers are not taken into account. Their analysis focused on the application of boarding strategies in conventional cabins. TOMICS by the German Aerospace Center [87] is a DES built for analyzing passenger flows within airport facilities, such as terminal areas and security checks, as well as boarding and disembarking processes. The simulation features a comprehensive path-finding algorithm and cabin procedures, like luggage stowing. The commercial CAST Cabin framework [63] is based on an ABS approach and provides a promising implementation to model passenger boarding processes. A similar approach was pursued by Richter [88] featuring a variety of agent properties as well as executable sub-tasks, such as luggage stowing. Unfortunately, this framework suffered from performance issues with an increasing number of agents. The approach of Fuchte [32] was based on the latter model and eliminates performance issues. The simulation includes an overhead bin model considering the actual remaining luggage capacity. The implemented path finding algorithm disregarded other agents which leads to queuing of passengers and during the simulation, no dynamic change of the path is possible. A patent from

[6] NetLogo is available under the terms of the GNU General Public License (GPL). Further information can be obtained from https://ccl.northwestern.edu/netlogo/

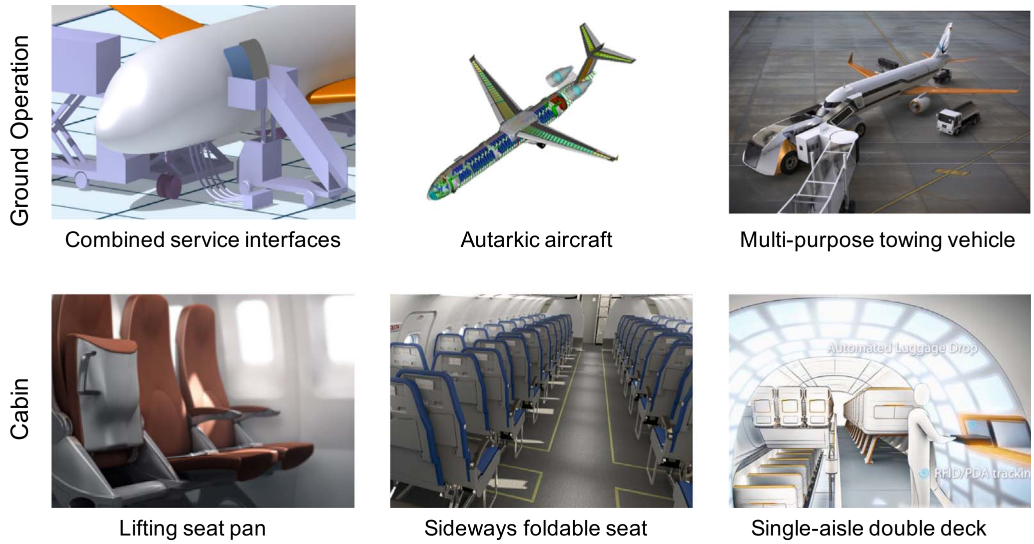


**Fig. 8.** Overview of concepts for aircraft ground operation and cabin modifications [11,95,96,110,111,114].

Boeing [89] described a 3D ABS boarding and disembarking tool. Passenger characteristics were described with walking speed and amount of HL, and their interferences are modeled with a predefined waiting time. The open-source passenger flow simulation framework PAXelerate[7] by Schmidt et al. [8,10,90] is based on a microscopic approach applying agent-based modeling techniques. This framework targets the assessment of novel cabin architectures through the modeling of agent interactions and detailed HL stowing.

## 5. Review of ground operation research projects and conceptual studies

Recent research projects and conceptual studies are targeting an evolution of the ground operation processes towards improved time efficiency, increased predictability, and reduced disruptions. The targeted time frame spans from short-term concepts ( < 10 years) to long-term disruptive solutions ( > 30 years). The section first summarizes studies which focus on the aircraft turnaround event before pathways for improved passenger processes are explored. The presented study results were produced using the agent-based passenger flow simulation framework PAXelerate. A selection of the covered concepts is depicted in Fig. 8.

### 5.1. Aircraft ground operation concepts and studies

A review of research projects and conceptual studies covers ground operational concepts in terms of airport infrastructure, passenger and turnaround processes and aircraft conceptual changes. From an airport point of view, modifications of current boarding bridges were investigated to speed up passenger processes using multiple doors [91]. An additional underground supply system for electrical power, potable and waste water and PCA, besides the existing fuel pipe system, could minimize the number of GSE and apron traffic [92–94]. A relocation of the aircraft interfaces for fuel, potable and waste water, ground power and PCA would become feasible in this context. Process times were unaffected by the changes, however, the number of required vehicles was dramatically reduced as well as required manpower and complexity of the ground servicing procedures. The aircraft itself would need additional ducts and pipes if traditional locations of sub-systems are not changed [11]. The application to existing airports is limited for most of these concepts, since infrastructural modifications would disrupt the operation heavily and require a huge investment to change. Furthermore, backwards compatibility with current aircraft in-service should be ensured.

On the road to more automated apron services, automated de-icing and GSE docking could be accomplished in the short term. A combination of machine vision algorithms and targets on the aircraft fuselage would enable the automated docking of passenger boarding bridges or cargo loaders [91]. A further step would be fully autonomous moving ground vehicles with remote monitoring and control. In this context, robots would be used to transfer baggage, consumables and equipment [91–94]. These concepts could minimize the GSE docking and maneuvering time while simultaneously avoiding potential collisions. Mature technologies would also be required to provide security against hacking of the control system and in case of incidents, liability issues would have to be addressed.

In the past, studies have investigated aircraft which were equipped with all tools and equipment required for an autarkic turnaround [92–95] without relying on GSE. Novel aircraft configurations that integrate innovative subsystems optimized for ground handling activities would increase the independence from airport resources and reduces the number of required GSE. A quantitative assessment of high-wing aircraft with tail-mounted engines, integrated stairs, foldable seats and conveyor belts inside the compartment revealed around 4% higher DOC per seat-mile compared to a state-of-the-art reference aircraft. The gained benefit of the increased utilization could not outweigh the drawback of a higher maximum take-off weight due to the embedded systems [95].

The so-called SkyGate vehicle [96–98], which was designed with a minimal number of interfaces between aircraft and ground should reduce turnaround time through the realization of passenger egress and ingress during taxiing operations. The vehicle combined the functions of a towing truck, a shuttle bus and a jetbridge into a single multi-functional vehicle driven by electric motors and controlled autonomously. The main idea behind this concept was to let passengers de-board the aircraft directly into the new vehicle as soon as it was docked to the aircraft, directly after the latter has vacated the runway upon arrival. While the disembarking is in progress, SkyGate tows the plane to its parking position, allowing the aircraft engines to be switched off for fuel-saving and noise-reduction purposes. After completing the disembarking process and separating from the aircraft at its parking position, the vehicle continues its journey to the terminal, where passengers can disembark. A first assessment showed an improvement potential of 8% in reduced fuel consumptions per flight

[7] The source code of the agent-based passenger flow simulation "PAXelerate" and any accompanying materials are available under the terms of the Eclipse Public License (EPL) v1.0. Further information can be obtained from http://www.paxelerate.com

and increased annual aircraft utilization by 11%. However, regulatory barriers concerning the passenger movement during taxiing still exist [96–98].

### 5.2. Studies and concepts to improve passenger processes

In literature, five general development directions for improved passenger processes and advanced aircraft cabins can be identified: boarding schemes, aisle, door, seats and general layout modifications. An overview of the associated concepts is listed in Table 3 and key aspects are highlighted in the following [9].

#### 5.2.1. Boarding strategies

Airlines seek alternative strategies to increase boarding process efficiency. They have applied various boarding strategies that call for a predefined sequence of passengers entering the cabin depending on their allocated seat. Nyquist [99] and recently Jaehn and Neumann [100] provided an overview of Operation Research in this field covering methods such as back-to-front, zone boarding or random seating. Steffen and Hotchkiss [101,102] conducted an experimental comparison of different airplane boarding methods and found a significant reduction in the boarding times when using improved methods. Methods which call for parallel boarding processes showed more efficient use of the aisle. If more passengers stowed their luggage simultaneously, the number of aisle interferences was reduced, which lead to increased speed in passenger ingress by almost 25%. Building upon these findings, Milne and Kelly [103] presented a modified method in which passengers were individually assigned seats based on the amount of luggage they carry with the goal of distributing them evenly throughout the aircraft. The results showed efficiency gains of up to 28%. However, most of the strategies tested are not practical in regular flight operation, as passengers must be grouped in a predefined way, which can split some groups. To avoid that, Zeineddine [104] proposed a dynamically optimized boarding strategy where passengers were sequenced in a boarding queue based on their seats' positions, associated groups, and the possibility of interferences, immediately after the last check-in. The results of this strategy showed similar performance compared to the Steffen strategy with 24% lower boarding times [102].

Only a few studies have investigated disembarking methods. The results showed that the commonly used random strategy is competitive, but could be enhanced by a reverse window-to-aisle strategy where passengers seated on the aisle leave the aircraft first [29,105,106]. In this case, the applicability in regular operation is questionable due to the intrinsic passenger urge to leave the plane.

#### 5.2.2. Aisle

A widening of the aisle results in a reduced seat width or in an increased aircraft cross-section diameter. The latter requires new aircraft design programs which manufactures try to avoid due to high development and certification expenses. The minimum aisle width for passenger aircraft is defined as 0.38 m (15 in.) on the floor and 0.51 m (20 in.) at 0.64 m (25 in.) above the floor level (CS/FAR 25.815) [17,18]. In current SA configurations, the aisle width is between 0.48 and 0.64 m (19–25 in.) [20,107]. First studies of an aisle widening of 0.2 m (8 in.) showed a boarding time reduction potential for common SA aircraft, of 5–7%, depending on the cabin size and number of passengers. This resulted from fewer aisle interferences due to an increased number of overtaking possibilities [32].

Switching from a SA layout to a TA configuration allows for a passenger flow separation into two different streams. This shortens the queue lengths and number of aisle interferences. However, traditional SA configurations are superior in drag, weight and fuel burn from an aircraft design point of view. Fuchte [32] and Schmidt et al. [9] investigated single- and twin-aisle cabin layouts in the range from 150 to 340 seats. A patent from Boeing [108] showed a concept for around 200 passengers in a TA configuration, preferably, with a seven-abreast configuration. A seven-abreast TA out-performed a six-abreast SA with 180 seats in terms of boarding time by 40–50% depending on the HL degree, as illustrated in Fig. 9. This resulted from fewer seat interferences, slightly reduced walking distances and added overhead volume due to a larger cross-section.

#### 5.2.3. Door

Changes to the door positions from the cabin ends to the middle allow for the passenger flow division into two separate streams. For SA aircraft above 180 seats, a so-called quarter door could achieve a 3–24% boarding time reduction. A quarter door (L2) is placed at around 25% of the fuselage length in front of the wing where a three-quarter door (L3) is located at 75% after the wing root, as illustrated in Fig. 10. The results, correlated with the fuselage length, show larger potentials for higher passenger numbers due to the separation of the doors along the fuselage. For smaller aircraft, the limited fuselage length does not grant a sufficient margin between quarter and forward doors. This concept, however, has a substantial aircraft weight penalty below 220 seats, since then no full-size emergency exits are required [9,32]. Similar gains of 26% have been shown by Schultz et al. [30] in an A320 with 174 seats. Airbus and Boeing have presented in their recent derivatives of the A320 and B737 family that it is possible to implement minor modifications to doors and emergency exits enabling larger seating capacity and improved cabin layouts [33,34]. Thus, an enhanced aircraft could be offered to airlines without engaging in the extensive certification process when modifications to the cross-section are performed.

Significant improvements could be made using the front (L1) and the rear door (L4) simultaneously for passenger processes (see Fig. 11). Fuchte [32] and Schmidt et al. [9] showed a 30–35% boarding time

**Table 3**
Overview of cabin modifications.

| | Concept | Benefit | Method | Ref. |
|---|---|---|---|---|
| Strategy | Steffen and Hotchkiss | 25%[a] | simulation | [101,102] |
| Strategy | Milne and Kelly | 28%[a] | simulation | [103] |
| Strategy | Zeineddine | 24%[a] | simulation | [104] |
| Aisle | Aisle width (0.2 m) | 5–7% | simulation | [32] |
| Aisle | Multi-aisle (two) | 40–50% | simulation | [10,32] |
| Door | Quarter door | 3–24% | simulation | [10,30,32] |
| Door | Door size | – | – | – |
| Door | Number of doors (two) | 30–47% | simulation | [10,32] |
| Seat | Sideways foldable seat (SFS) | 28–47% | simulation | [9,10,109–113] |
| | Lifting seat pan (LSP) | | | |
| Layout | Increased seat pitch | – | – | – |
| Layout | Multi-deck | – | – | [96,116] |

[a] Benefits are measured against random boarding [9].

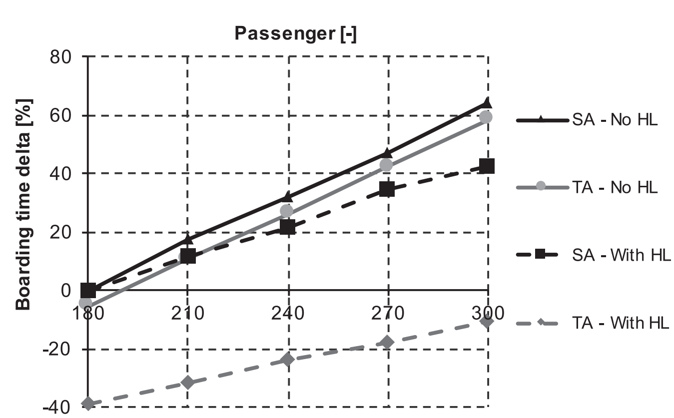


**Fig. 9.** Boarding times for single- (SA) and twin-aisle (TA) configurations using one door with and without hand luggage (HL) [9].

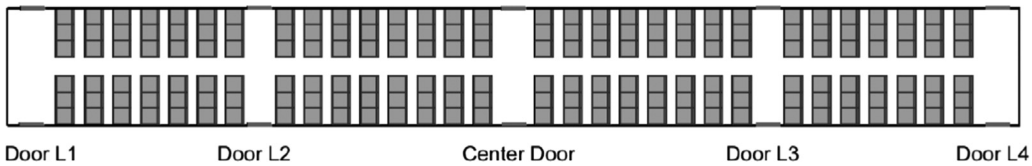


**Fig. 10.** Exemplary depiction of door positions for a single-aisle six abreast configuration.

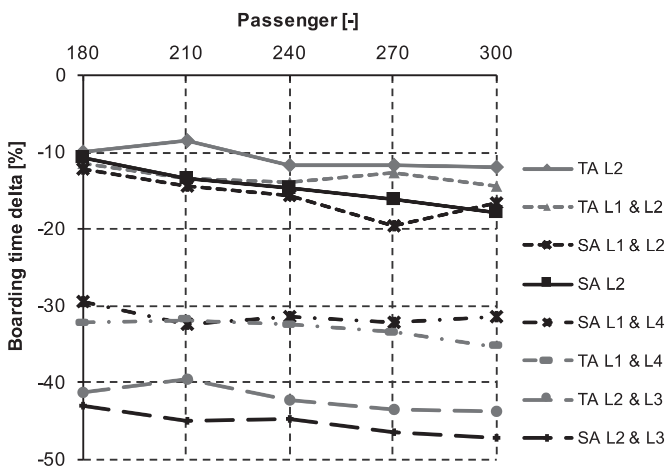


**Fig. 11.** Influence of door position permutations on the passenger boarding times for single-(SA) and twin-aisle (TA) aircraft [9].

reduction for this scenario. Better performance could be accomplished through the combination of a quarter (L2) and three-quarter door (L3) as proposed by Schmidt et al. [11]. This concept shows reduced ingress times for the TA of 40–43% and for the SA aircraft of 43–47%, as illustrated in Fig. 11. A center door (CD) performed slightly better than the quarter door (L2) for both cases where a center door allowing two passengers simultaneously enter the cabin (CD −2 PAX) achieved 30–37% lower boarding times. This larger door would split the passengers into two streams within the jetway.

### 5.2.4. Seat

On-demand adaptable cabins during boarding promise significant reductions in passenger egress and ingress times. The concept of a lifting seat pan (LSP), also referred to as cinema seats (Fig. 12 left), was introduced by AIDA Development [109] and initially assessed by Hertl [110]. The aim is to increase the moving space of passengers in the row and enhance their access to the overhead bins. The foldable seat pan allows passengers to step into the row, if the aisle seat is not yet occupied, and to stow their HL in the overhead bin without blocking the aisle. If passengers try to get to their window or middle seat when the aisle seat is occupied, the aisle passengers can stand up while remaining within the row, reducing the duration of aisle interferences. A similar concept called sideways foldable seat (SFS) allows the aisle seat to slide over the middle seat, as proposed by Molon [111], or under the middle seat, as introduced by Isikveren et al. [112,113] (see Fig. 12 right). This enables a three-fold increase of the aisle width allowing passengers to seamlessly pass other passengers who stow their HL in the overhead bins.

Recent studies by Schmidt et al. [9,10] showed the potential impact of foldable seat concepts during passenger boarding. Fig. 13 illustrates the results for SA aircraft configurations accommodating 180–300 passengers. Efficiency gains between 28% and 47% could be determined taking a layout with conventional seats as reference. The shortest boarding times were observed for 180 passengers using two doors. However, the operational applicability of the concepts relies on the certification and passenger acceptance in terms of manageability, integration of inflight amenities and seating comfort. The folding mechanism introduces more complexity which leads to higher maintenance efforts and weight [113]. A removal of the metal strap prohibiting HL to move around the cabin would affect the stowage of luggage under the seat, which is a major drawback with current high amounts of hand luggage. Furthermore, the seat belt would require a redesign to prevent getting blocked [10].

### 5.2.5. Cabin layout

Although beneficial for passenger comfort, an increased seat pitch is disadvantageous for airlines as it reduces the cabin's capacity. For boarding and disembarking procedures, a sufficient seat pitch would enable passengers to get to their seat without the need of other passengers to stand up, thereby eliminating row interferences. Furthermore, passengers could stow their luggage in the overhead bins without blocking the aisle, significantly reducing aisle interferences.

Three different seat abreast variations between six and eight were examined for TA cabins. Fig. 14 depicts the boarding time deviation with the TA six-abreast case as datum. A larger seat abreast resulted in higher boarding times and no distinct dependency on the cabin size was identified. The seven-abreast configuration showed 4–7% longer ingress times and the eight-abreast 9–17%.

An alternative proposal is the arrangement of passengers on two decks which could increase the number of seats considerably. A design

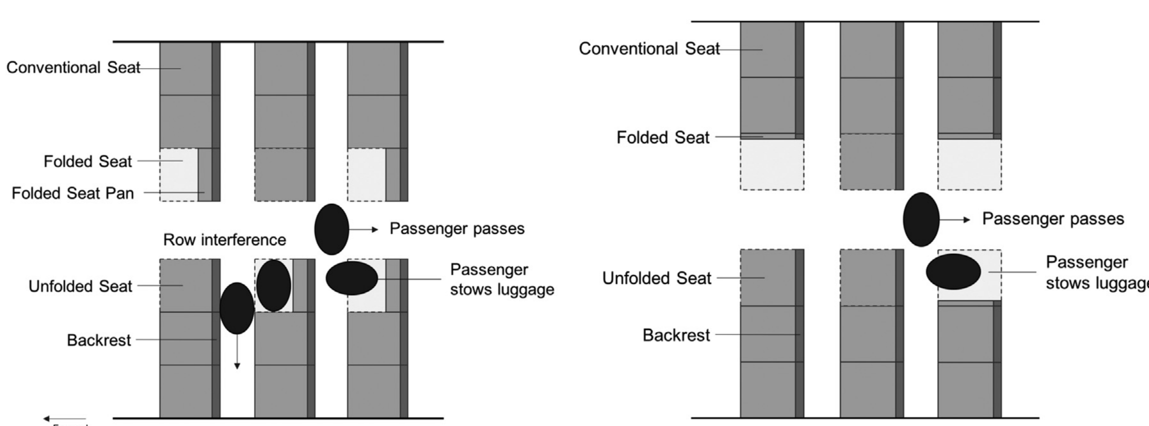


**Fig. 12.** Lifting seat pan (LSP) concept (left) and sideways foldable seat (SFS) concept (right) in a six-abreast arrangement with folded and unfolded seats [9,10].

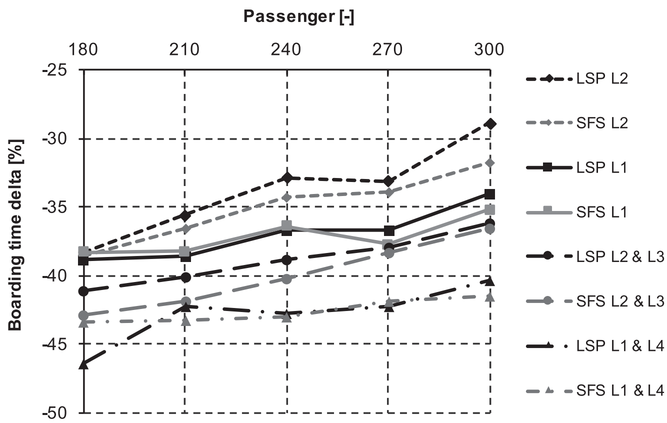


**Fig. 13.** Boarding performance for single-aisle aircraft with LSP and SFS compared to conventional seats under the consideration of hand luggage (based on [10]).

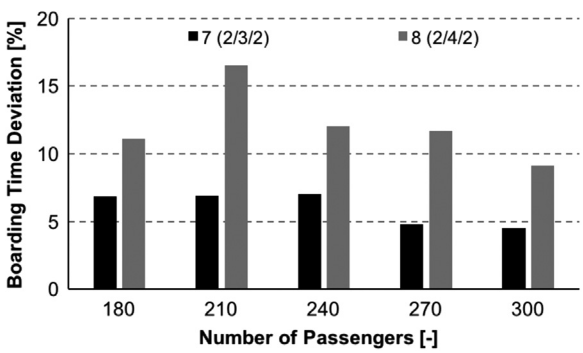


**Fig. 14.** Impact of the seat abreast for twin-aisle configurations (baseline: six-abreast twin-aisle (TA6)).

study, based on a typical narrow-body aircraft, accommodates several passengers in the under-floor space, which is used today for cargo [96,114]. This change would require only a slight enlargement of the fuselage diameter. Rearranging the aircraft doors would enable parallel passenger boarding on the lower and upper deck, as well as, on both fuselage sides. First estimations showed that current egress and ingress time could be retained despite a 20% increase in number of passengers [96].

## 6. Recapitulation

The ground operational processes are the connecting element between aircraft en-route operations and airport infrastructure. Aircraft operators are challenged by prolonged passenger process times, airport capacity constraints, schedule disruptions, reduced aircraft utilization and the increasing cost pressure. Therefore, industry and research organizations are looking for solutions to enable competitive aircraft operations for all involved stakeholders.

The assessment of novel ground operational concepts relies on the modeling and simulation of the processes and interaction during the early stages of the product development. The variety of models and frameworks dealing with the aircraft turnaround cover resource allocation, operational uncertainties, as well as, the detailed modeling of each sub-process. In the field of simulating the passenger processes inside of the cabin, a variety of macroscopic and microscopic models exist. Especially, DES and ABS approaches allow for the modeling of the individual behavior of each passenger during the boarding and disembarking process. Unfortunately, most of the reviewed frameworks are only published partly in form of research papers and do not allow the access of the model foundation which is required for a deeper understanding and model extension. Here, the open-source agent-based passenger flow simulation framework PAXelerate developed by the author was applied to assess novel aircraft cabin concepts.

The review of conducted studies and developed concepts showed opportunities to free up airport capacity through an optimized use of airport infrastructure and the reduction of additional buffers allowing faster door-to-door travel for passengers which could generate higher revenues for airlines. Promising approaches try to reduce apron traffic through underground supply of liquids and parallelize passenger processes and taxiing. The regulated-driven sequential order of passenger, cabin processes and refueling often constitutes the critical path. Thus, a further parallelization or shortening of these processes would reduce the overall turnaround increasing the aircraft utilization. However, this could push other resources to its limits, such as gate positions and taxiways, and so far unconstrained processes might become constrained, including aircraft brake cooling times or cockpit procedures.

One aspect which cannot be influenced by the stakeholders are the passengers and their behavior patterns. Not only would airlines profit from shorter boarding procedures, since it is the last process before the departure but also each passenger. The application of boarding strategies promises significantly lower ingress times but often disregards operational facts, such as HL distribution or passenger group behavior. Thus, novel cabin layouts focusing on aisle, door, seat and general layout modifications are another way to increase boarding efficiency. Alternative door positions and adaptable seats show large efficiency potentials with over 40% lower passenger process times with a comparatively low market entry barrier. However, the operational applicability of the concepts integration still has to be shown.

## 7. Outlook

The review of concepts dealing with aircraft ground operation showed several promising approaches to increase the current efficiencies, however, the variety and concept maturity is manageable. During the development of ideas and concepts, researchers should rethink the current practice when it concerns the aircraft ground operation and target a seamless passenger journey. Future research should focus on the further elaboration of these concepts in terms of their operational robustness, technology maturity and certification. The latter requires to initiation of dialogue with national and international regulators to clear the way for an envisaged concept entry into service. Furthermore, a holistic assessment during the early stages of the preliminary aircraft design process would allow the identification of benefits and drawbacks for all involved stakeholders. These findings are often the basis for decision making for further resource allocation pursuing the overall goal of reducing the operating cost.

## Acknowledgements

The authors would like to express their gratitude to Munich Aerospace e.V. for their support and Christoph Schinwald, Philipp Heinemann and Kay Plötner for fruitful discussions and valuable advice.

## References

[1] Airbus, Global Market Forecast - Flying by Numbers, Technical report, 2015.
[2] Boeing, Current Market Outlook 2015–2034, Technical report, 2015.
[3] NASA, Advanced Air Vehicle Program, 2016.
[4] Air Transport Action Group (ATAG), The Global Flightplan, in: Proceedings of the 19th Conference of the Parties (COP19) to the United Nations Framework Convention on Climate Change, updated with the results of the 38th ICAO Assembly, number November, 2013.
[5] Advisory Council for Aeronautical Research in Europe (ACARE), European Aeronautics: A Vision for 2020, Technical Report January 2001.
[6] Advisory Council for Aviation Research and Innovation in Europe (ACARE), Strategic Research & Innovation Agenda (SRIA) - Volume 1, Technical report, Brussels, 2012.
[7] Advisory Council for Aviation Research and Innovation in Europe (ACARE), Strategic Research & Innovation Agenda (SRIA) - Volume 2, Technical report,

Brussels, 2012.
[8] M. Schmidt, M. Engelmann, T. Brügge-Zobel, M. Hornung, M. Glas, PAXelerate - An Open Source Passenger Flow Simulation Framework for Advanced Aircraft Cabin Layouts, in: 54th AIAA Aerosp. Sci. Meet., American Institute of Aeronautics and Astronautics. doi: 10.2514/6.2016-1284, 2016.
[9] M. Schmidt, M. Engelmann, R. Rothfeld, M. Hornung, Boarding Process Assessment of Novel Aircraft Cabin Concepts, in: Proceedings of the 30th International Congr. Aeronaut. Sci. - ICAS, Daejoun, 2016.
[10] M. Schmidt, P. Heinemann, M. Hornung, Boarding and Turnaround Process Assessment of Single- and Twin-Aisle Aircraft, in: Proceedings of the 55th AIAA Aerosp. Sci. Meet., Dallas, Texas, USA. doi: 10.2514/6.2017-1856, 2017.
[11] M. Schmidt, P. Nguyen, M. Hornung, Novel Aircraft Ground Operation Concepts Based on Clustering of Interfaces, in: Proceedings of the SAE International, Seattle, USA. http://dx.doi.org/10.4271/2015-01-2401, 2015.
[12] R. Callahan, Impact of the Next Two Aircraft Generations on Airport Design, in: Proceedings of the AIAA 7th Annual Meeting and Technical Display, Houston, Texas, 1970.
[13] R.W.K. Leung, J.W. Little, D.C.H. Li, Getting Ready for the A380 Aircraft at Hong Kong International Airport, in: Proceedings of the 2007 FAA Worldwide Airport Technology Transfer Conference, number April, Atlantic City, New Jersey, USA, 2007.
[14] M. Schmidt, A. Paul, M. Cole, K.O. Ploetner, Challenges for ground operations arising from aircraft concepts using alternative energy, J. Air Transp. Manag. 56 (2016) 107–117. http://dx.doi.org/10.1016/j.jairtraman.2016.04.023.
[15] N.J. Ashford, H.P.M. Stanton, C.A. Moore, P. Coutu, J.R. Beasley, Airport Operations, 3rd ed., McGraw-Hill, 2013.
[16] International Air Transport Association (IATA), Airport Handling Manual (AHM) −34th Edition, Technical report, 2013.
[17] European Aviation Safety Agency (EASA), Certification Specifications for Large Aeroplanes CS-25, 2009.
[18] Federal Aviation Administration (FAA), Federal Aviation Regulations (FAR) Part 25 - Airworthiness Standards: Transport Category Airplanes, 2015.
[19] A.H. Midki, R.J. Hansman, T.G. Reynolds, Air carrier flight operations, Technical Report, July 2004.
[20] Airbus, A320 - Aircraft Characteristics Airport and Maintenance Planning, Technical report, 2015.
[21] H. Fricke, M. Schultz, Delay Impacts onto Turnaround Performance, in: Proceedings of the USA/Europe Air Traffic Manag. Res. Dev. Semin., Napa, USA, 2009.
[22] C.-L. Wu, Monitoring aircraft turnaround operations – Framework development, application and implications for airline operations, Transp. Plan. Technol. 31 (2008) 215–228. http://dx.doi.org/10.1080/03081060801948233.
[23] S. Marelli, G. Mattocks, R. Merry, The role of computer simulation in reducing airplane turn time, Aero Mag. 1 (1998).
[24] NEA, Survey on standard weights of passengers and baggage, Technical report, Zoetermeer, 2009.
[25] U.S. Department of Health and Human Services, Anthropometric Reference Data for Children and Adults: United States, 2007–2010, Technical Report 252, 2010.
[26] N. Easen, Business travel gets a female touch, BBC Travel - Passpt. Blog. (2013). ⟨http://www.bbc.com/travel/blog/20130213-business-travel-gets-a-female-touch⟩.
[27] A. Strässle, Dutch Vision, NeueZürcher Zeitung, ⟨http://www.nzz.ch/aktuell/startseite/hollaendischer-weitblick-1.549440#⟩. (originally in German), 2007.
[28] B. Yuan, J. Yin, M. Wang, STAR: (saving time, adding revenues) boarding / deboarding strategy, UMAP J. 28 (2007) 371–384.
[29] H.S. Appel, Analysis of aircraft boarding delays and investigation their mitigation (Dissertation), RWTH Aachen University, 2014 (originally in German).
[30] M. Schultz, T. Kunze, H. Fricke, Boarding on the critical path of the turnaround, in: Proceedings of the Tenth USA/Europe Air Traffic Manag. Res. Dev. Semin. Board., 2013.
[31] European Comission, Comission Regulation (EC) No 859/2008, Technical Report July 2008.
[32] J. Fuchte, Enhancement of Aircraft Cabin Design Guidelines with Special Consideration of Aircraft Turnaround and Short Range Operations (Dissertation), Technische Universität Hamburg-Harburg, 2014.
[33] Airbus, Airbus offers added seating capacity for the A320 Family while retaining modern comfort standards, Press Release. (2014). ⟨http://www.airbus.com/presscentre/pressreleases/press-release-detail/detail/airbus-offers-added-seating-capacity-for-the-a320-family-while-retaining-modern-comfort-standards/⟩ (accessed 19 February 2015).
[34] Boeing, Boeing Launches 737 MAX 200 with Ryanair, Press Release. (2014). ⟨http://boeing.mediaroom.com/2014-09-08-Boeing-Launches-737-MAX-200-with-Ryanair⟩ (accessed 19 February 2015).
[35] M. Schmidt, K.O. Plötner, C. Pornet, A.T. Isikveren, M. Hornung, Contributions of Cabin Related and Ground Operation Technologies Towards Flightpath 2050, in: Proceedings of the Dtsch. Luft- Und Raumfahrtkongress 2013, Stuttgart, 2013.
[36] MIT, Airline Data Project, ⟨http://web.mit.edu/airlinedata/www/default.html⟩, 2016.
[37] C. Schinwald, M. Hornung, Methodical Approach to Determining the Capacity Utilisation of Airports: the Development of the European AirTraffic System between 2008 and 2013, in: Proceedings of the Dtsch. Luft- Und Raumfahrtkongress 2014, Augsburg, Germany, 2014.
[38] H. Balakrishnan, I. Deonandan, I. Simaiakis, Opportunities for Reducing Surface Emissions Through Surface Movement Optimization, MIT International Cent. Air Transp, 2008.
[39] Eurocontrol, CODA - Delays to Air Transport in Europe - Annual 2014, Technical report, 2014.
[40] S. AhmadBeygi, A. Cohn, M. Lapp, Decreasing airline delay propagation by re-allocating scheduled slack, IIE Trans. 42 (2010) 478–489. http://dx.doi.org/10.1080/07408170903468605.
[41] M. Schmidt, The five minute turnaround – a reasonable goal? An overview of ongoing ground operations research, invited talk at in: Proceedings of the 9th Asian GHI Conference, 2016.
[42] M. Mirza, Economic impact of airplane turn-times, Boeing Aero Mag. (2008) 15–19.
[43] D. Gillen, A. Lall, Competitive advantage of low-cost carriers: some implications for airports, J. Air Transp. Manag. 10 (2004) 41–50. http://dx.doi.org/10.1016/j.jairtraman.2003.10.009.
[44] A.M. Pilarski, Why Can't We Make Money in Aviation?, Ashgate Publishing Ltd., Hampshire, England, 2007.
[45] E. Clayton, A. Hilz, Industry perspectives −2015 Aviation Trends, 2015.
[46] International Air Transport Association (IATA), Airline Cost Performance, Technical report, 2006.
[47] F.L. Levy, G.L. Thompson, J.D. Weist, The ABCs of the critical path method, Harv. Bus. Rev. 41 (1963) 98–108.
[48] B. Page, N. Knaaka, S. Kruse, A discrete event simulation framework for agent-based modelling of logistic systems, in: Inform. 2007 Inform. Trifft Logistik: pp. 397–404, 2007.
[49] E. Bonabeau, Agent-based modeling: methods and techniques for simulating human systems, PNAS 99 (2002) 7280–7287. http://dx.doi.org/10.1073/pnas.082080899.
[50] J.P. Braaksma, J.H. Shortreed, Improving airport gate usage with critical path, Transp. Eng. J. ASCE 97 (1971) 187–203.
[51] W.H. Ip, V. Cho, N. Chung, G. Ho, A multi agent based model for Airport service planning, Int. J. Eng. Bus. Manag. 2 (2010) 1. http://dx.doi.org/10.5772/9718.
[52] A. Vidosavljević, V. Tošić, Modeling of turnaround process using petri nets, in: Proceedings of the 14th ATRS World Conference, 2010.
[53] P.G. Voulgarellis, M.A. Christodoulou, Y.S. Boutalis, A MATLAB Based Simulation Language for Aircraft Ground Handling Operations at Hub Airports (SLAGOM), in: Proceedings IEEE International Symp. Mediterrean Conference Control Autom. Intell. Control. http://dx.doi.org/10.1109/.2005.1467037, 2005.
[54] G. Ioannidis, M.A. Christodoulou, Y.S. Boutalis, A SIMULINK Based Simulation Language for Aircraft Simulation of Ground Handling Operations at Hub Airports (SSLA), in: Proceedings IEEE International Symp. Mediterrean Conference Control Autom. Intell. Control. http://dx.doi.org/10.1109/.2005.1467038, 2005.
[55] K. Andersson, F. Carr, E. Feron, W.D. Hall, Analysis and Modeling of Ground Operations at Hub Airports, in: Proceedings of the 3rd USA/Europe Air Traffic Manag. R & D Seminar: pp. 13–16. http://dx.doi.org/10.1109/JSTSP.2013.2237882, 2000.
[56] K. Andersson, Potential Benefits of Information Sharing During the Arrival Process At Hub Airports (Master Thesis), MIT, 2000.
[57] C. Wu, R.E. Caves, Modelling and simulation of aircraft turnaround operations at airports, Transp. Plan. Technol. 27 (2004) 25–46. http://dx.doi.org/10.1080/0308106042000184445.
[58] C. Wu, R.E. Caves, Modelling and optimization of aircraft turnaround time at an airport, Transp. Plan. Technol. 27 (2004) 47–66. http://dx.doi.org/10.1080/0308106042000184454.
[59] M.M. Mota, N. De Bock, G. Boosten, E. Jimenez, J. Pinho, Simul.-Based turnaround Eval. Lelystad Airpt. (2015).
[60] A. Norin, T.A. Granberg, D. Yuan, P. Värbrand, Airport logistics - A case study of the turn-around process, J. Air Transp. Manag. 20 (2012) 31–34. http://dx.doi.org/10.1016/j.jairtraman.2011.10.008.
[61] Rockwell Automation, Arena simulation software, 2016.
[62] Y. Tian, H. Liu, H. Feng, B. Wu, G. Wu, Virtual Simulation-Based Evaluation of Ground Handling for Future Aircraft Concepts, J. Aerosp. Inf. Syst. 10 (2013) 218–228. http://dx.doi.org/10.2514/1.49836.
[63] Airport Research Center, CAST Cabin, 2016.
[64] M. Bevilacqua, F.E.E. Ciarapica, G. Mazzuto, C. Paciarotti, The impact of business growth in the operation activities: a case study of aircraft ground handling operations, Prod. Plan. Control. 26 (2015) 564–587. http://dx.doi.org/10.1080/09537287.2014.939234.
[65] ProSim, ProSimPlus - Steady-state simulation and optimization of processes (2016), ⟨http://www.prosim.net/⟩.
[66] Lanner, WITNESS - predictive simulation software for modelling and application development, ⟨http://www.lanner.com/technology/witness-simulation-software.html⟩.
[67] H. Fricke, M. Schultz, Improving Aircraft Turn Around Reliability, in: Proceedings of the Third International Conference Res. Air Transp: pp. 335–343, 2008.
[68] B. Oreschko, M. Schultz, J. Elflein, H. Fricke, Significant TurnaroundProcess Variations due to Airport Characteristics, in: Proceedings of the Air Transp. Oper. Symp, 2010.
[69] A. Schlegel, Turnaround processes in aviation: a statistical analysis using the example of the German Lufthansa AG at Frankfurt airport (Dissertation), Georg-August-Universität Göttingen, Germany, 2010. http://dx.doi.org/10.1007/978-3-8349-8691-7_2 (originally in German).
[70] D.R. Sanchez, Analysis of Ground Handling Characteristics of Innovative Aircraft Configurations (Master Thesis), Hamburg University of Applied Science, 2009.
[71] E. Bachmat, D. Berend, L. Sapir, S. Skiena, N. Stolyarov, Analysis of aeroplane boarding via spacetime geometry and random matrix theory, J. Phys. A. Math. Gen. 39 (2006) L453–L459. http://dx.doi.org/10.1088/0305-4470/39/29/L01.
[72] E. Bachmat, D. Berend, L. Sapir, S. Skiena, N. Stolyarov, Analysis of airplane boarding Times, Oper. Res. 57 (2009) 499–513. http://dx.doi.org/10.1287/

opre.1080.0630.
[73] M.H.L.VanDen Briel, J.R. Villalobos, G.L. Hogg, The Aircraft Boarding Problem, in: Proceedings of the 12th Ind. Eng. Res. Conference, 2003.
[74] M.H.L. Van Den Briel, J.R. Villalobos, G.L. Hogg, T. Lindemann, A.V. Mulé, America west airlines develops efficient boarding strategies, Interfaces (Provid.). 35 (2005) 191–201. http://dx.doi.org/10.1287/inte.1050.0135.
[75] M. Bazargan, A linear programming approach for aircraft boarding strategy, Eur. J. Oper. Res. 183 (2007) 394–411. http://dx.doi.org/10.1016/j.ejor.2006.09.071.
[76] M. Soolaki, I. Mahdavi, N. Mahdavi-Amiri, R. Hassanzadeh, A. Aghajani, A new linear programming approach and genetic algorithm for solving airline boarding problem, Appl. Math. Model. 36 (2012) 4060–4072. http://dx.doi.org/10.1016/j.apm.2011.11.030.
[77] H. Van Landeghem, a. Beuselinck, Reducing passenger boarding time in airplanes: a simulation based approach, Eur. J. Oper. Res. 142 (2002) 294–308. http://dx.doi.org/10.1016/S0377-2217(01)00294-6.
[78] P. Ferrari, K. Nagel, Robustness of efficient passenger boarding strategies for airplanes, Transp. Res. Rec. (1915) 44–54. http://dx.doi.org/10.3141/1915-06.
[79] R. Livermore, W. Hall, M. Keynes, A multi-agent system approach to a simulation study comparing the performance of aircraft boarding using pre-assigned seating and free-for-all strategies. 2008.
[80] J. Audenaert, K. Verbeeck, G.Vanden Berghe, Multi-agent based simulation for boarding, in: Proceedings of the 21st Benelux Conference Artif. Intell, 2009.
[81] R. Cimler, E. Kautzká, K. Olševičová, M. Gavalec, Agent-based model for comparison of aircraft boarding methods, in: Proceedings of the 30th International Conference Math. Methods Econ. Agent-Based 73–78, 2012.
[82] U. Wilensky, NetLogo - a multi-agent programmable modeling environment, ⟨https://ccl.northwestern.edu/netlogo/⟩, 2016.
[83] S.J. Qiang, B. Jia, D.F. Xie, Z.Y. Gao, Reducing airplane boarding time by accounting for passengers' individual properties: a simulation based on cellular automaton, J. Air Transp. Manag. 40 (2014) 42–47. http://dx.doi.org/10.1016/j.jairtraman.2014.05.007.
[84] M. Schultz, Development of an agent-based model to investigate the passenger behavior in airport terminals (Dissertation), Technical University of Dresden, 2010 (originally in German).
[85] A. Steiner, M. Philipp, Speeding up the airplane boarding process by using pre-boarding areas boarding areas, in: Proceedings of the 9th Swiss Transp. Conference, Ascona, 2009.
[86] Fire safety engineering group, The Faculty of Architecture, Computing & Humanities, University of Greenwich AirExodus, (2015). ⟨http://fseg.gre.ac.uk/exodus⟩ (accessed 15 June 2015).
[87] Deutsches Luft- and Raumfahrt Zentrum, Traffic Oriented Microscopic Simulator – TOMICS. ⟨http://www.dlr.de/fw/en/desktopdefault.aspx/tabid-5980/9752_read-19750⟩ (accessed 15 June 2015), 2011.
[88] T. Richter, Simulationsmethodik zur Effizienz- und Komfortbewertung von Menschenflussprozessen in Verkehrsflugzeugen, Dissertation, Technical University ofMunich, 2007.
[89] E.R. Winkler, J. French, G.E. Mattocks, J.J. Konesky, Aircraft Passenger Flow System, Patent US 8,510,086 B1, 2013.
[90] M. Schmidt, M. Engelmann, R. Rothfeld, PAXelerate release alpha 0.7, Github Repos. (2016). http://dx.doi.org/10.5281/zenodo.56736.
[91] The INTERACTION Consortium, D3.6 Solutions Summary Report, Technical Report October 2014.
[92] The 2050+ Airport Consortium, D4.1 - The Time-Efficient Airport Concept - An operational concept for 2050 and beyond, Technical report, 2013.
[93] The 2050+ Airport Consortium. D4-2 The Ultra-Green Airport Concept - An operational concept for 2050 and beyond. Technical report, 2013.
[94] The 2050+ Airport Consortium. D4-3 The Cost-Effective Airport Concept - An operational concept for 2050 and beyond. Technical report, 2013.
[95] D. Scholz, P. Krammer, D.R. Sánchez, Final reportFHprofUnd-Projekt "ALOHA" – Aircraft design for reduced turn around cost, Hamburg, 2011. (originally in German).
[96] Bauhaus Luftfahrt e.V., Interdisciplinary cooperation in the research field of radical cabin architectures and novel ground handling processes. ⟨http://www.bauhaus-luftfahrt.net/archive/interdisziplinaere-zusammenarbeit-im-bereich-radikale-kabinenarchitekturen-und-neuartige-bodenabfertigungsprozesse?Set_language=en⟩, 2013.
[97] M. Stegmiller, Detailed Design of a Multifunctional Aircraft Ground Handling Vehicle (Term Thesis), Technical University of Munich, 2014.
[98] M. Schmidt, Handling Innovation - Tomorrows Turn?, Ground Handling International, 2015, pp. 42–43.
[99] D.C. Nyquist, K.L. McFadden, A study of the airline boarding problem, J. Air Transp. Manag. 14 (2008) 197–204. http://dx.doi.org/10.1016/j.jairtraman.2008.04.004.
[100] F. Jaehn, S. Neumann, Airplane boarding, Eur. J. Oper. Res. 244 (2014) 339–359. http://dx.doi.org/10.1016/j.ejor.2014.12.008.
[101] J.H. Steffen, J. Hotchkiss, Experimental test of airplane boarding methods, J. Air Transp. Manag. 18 (2012) 64–67. http://dx.doi.org/10.1016/j.jairtraman.2011.10.003.
[102] J.H. Steffen, Optimal boarding method for airline passengers, J. Air Transp. Manag. 14 (2008) 146–150. http://dx.doi.org/10.1016/j.jairtraman.2008.03.003.
[103] R.J. Milne, A.R. Kelly, A new method for boarding passengers onto an airplane, J. Air Transp. Manag. 34 (2014) 93–100. http://dx.doi.org/10.1016/j.jairtraman.2013.08.006.
[104] H. Zeineddine, A dynamically optimized aircraft boarding strategy, J. Air Transp. Manag. 58 (2017) 144–151. http://dx.doi.org/10.1016/j.jairtraman.2016.10.010.
[105] R. Cimler, K. Olševičová, Analysis Simulation of aircraft disembarking methods, in: Proceedings of the 3rd World Conference Inf. Technol., Barcelona, Spain: pp. 867–872, 2013.
[106] H. Appel, Application of aircraft disembarking strategies, in: Proceedings of the Dtsch. Luft- Und Raumfahrtkongress 2010, Hamburg, Germany, 2010. (originally in German).
[107] Boeing, B737 - Airplane Characteristics for Airport Planning. Technical report, 2013.
[108] M.M.K.V. Sankrithi, Twin aisle small airplane, Patent US 6,834,833 B2. doi:http://dx.doi.org/10.1016/j.(73), 2004.
[109] S.I.I. Group, Foldable Passenger Seat - The Aida way. ⟨http://www.sii-engineering.de/en/projects_passenger_seat⟩ (accessed 1 June 2016), 2016.
[110] M. Hertl, Assessment of an aicraft foldable seat conept in term of passenger movements, Technische Universität Berlin, 2005 (originally in German).
[111] Molon Labe Designs, Side-Slip Seat, (2015), ⟨http://www.airlineseats.biz⟩ (accessed 15 June 2015).
[112] A.T. Isikveren, A. Seitz, P.C. Vratny, C. Pornet, K.O. Plötner, M. Hornung, Conceptual Studies of Universally-Electric Systems Architectures Suitable for Transport Aircraft, in: Proceedings of the 61. Dtsch. Luft- Und Raumfahrtkongress 2012, Berlin, 2012.
[113] M. Götz, Engineering Concept Study of an Innovative Sideward Retractable Aircraft Seat (Term Thesis), Technical University of Munich, 2014.
[114] Bundesverband der Deutschen Luft- und Raumfahrtindustrie e.V. (BDLI), IMAGINE..., (2013). ⟨http://bdli.de/index.php?Option=com_content&view=article&id=2622⟩.